

Economics 753: Applied Econometrics – Fall 2019

Monday/Wednesday 10:30–11:45 & Lab Tuesday 9–10

Gordon Hall 303–304

<http://courses.umass.edu/econ753>

<http://courses.umass.edu/econ753/econ753.pdf>

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COURSE SYLLABUS

“There is no royal road to science, and only those who do not dread the fatiguing climb of its steep paths have a chance of gaining its luminous summits.”—Karl Marx

The purpose of this course is to help you become comfortable and creative as empirical economic researchers. We will therefore introduce a series of econometric techniques and models by observing life in the trenches, i.e., working through how practitioners have approached econometric problems and built models as vehicles for addressing substantive questions. We also put stress on communicating econometric results and descriptive statistics in ways that are illuminating, persuasive, and rigorous. But we will also ask the question: what does it mean to be rigorous with econometrics and statistics in general? Indeed, early in the course, we will even address so basic a methodological question as that raised by John Maynard Keynes himself in the infancy of econometrics—whether the entire enterprise was worth pursuing. Finally, in the course of your studying and then doing real-world econometrics, we aim to get you to become comfortable, and perhaps even proficient, in at least one major statistical software package and with manipulating datasets.

TEXTBOOKS AND READINGS

Most readings from the course are available from the course website or archives accessible to UMass students. Scanned chapters of *The Practice of Econometrics* (Ernest M. Berndt) are available from <http://courses.umass.edu/econ753/berndt.poe/Ch01.pdf> (substituting for 01 with other chapter numbers). We also recommend *Introduction to Econometrics* (James Stock and Mark Watson), *A Guide to Econometrics* (Peter Kennedy), and *Mostly Harmless Econometrics* (Joshua Angrist and Jörn-Steffen Pischke). Please bring the current readings to class.

REQUIREMENTS

There are two sets of requirements for the course:

1. **Problem sets.** We will assign six problem sets. You are required to complete all six, and the lowest grade will be dropped. We will not assign any exercises in the first week. After the first week, we will assign exercises every two weeks up to the last week of the course. The six exercises you turn in will collectively count for 50 percent of your final grade.
2. **Term paper.** The term paper assignment can be in one of two forms: a replication of an empirical econometric paper that interests you, or an original piece of econometric research. Given other demands over the course of the semester, we *strongly recommend* that you opt for the replication option, which should generally be less demanding than coming up with, estimating, and writing an original research paper. Replication can be an illuminating experience, especially if you choose a paper that fascinates you. Having said that, we will not stop anyone who is just burning with an original idea to get it down on paper. The term paper will count for the other 50 percent

of the grade. Some of the homework assignments will involve providing us with preliminary materials and updates on the term paper. Finally, we require all students to make oral presentations of their term paper on the last week of class. *We emphasize now, and will continue to do so during the term, the need to work steadily throughout the semester on this project.*

COMPUTER/DATA/SOFTWARE SUPPORT

Throughout the term we will conduct lab sessions to assist people in getting comfortable using a text editor and statistical application to manage and analyze data; we will focus on R, and advice on Stata and EViews is available. The instructors and TA will lead these lab sessions and will also hold regular office hours throughout the course to assist people further with their data and software needs and concerns.

COURSE STRUCTURE

I. Methodological Issues, Descriptive Statistics, and Visualization (3 classes)

Methodology

- J.M. Keynes, "Professor Tinbergen's Method," *Economic Journal*, September 1939, <http://www.jstor.org/stable/2224838>. *The Keynes piece is also reprinted in Keynes's collected works, Volume 14, pp. 306-317. Volume 14 of the collected works also contains other, more informal (and thus, in some ways, even more illuminating) writings by Keynes on his critique of Tinbergen.*
- Jan Tinbergen, "On a Method of Statistical Business-Cycle Research. A Reply." *Economic Journal*, March 1940, <http://www.jstor.org/stable/2225763>.
- J.M. Keynes, "On a Method of Statistical Business-Cycle Research. A Comment," *Economic Journal*, March 1940, <http://www.jstor.org/stable/2225764>.
- Wassily Leontief, "Academic Economics," *Science*, July 9, 1982, pp. 106-07, <http://www.jstor.org/stable/1689696>.
- Lawrence H. Summers, "The Scientific Illusion in Empirical Macroeconomics," *Scandinavian Journal of Economics*, 1991, <http://www.jstor.org/stable/3440321>.
- Deirdre N. McCloskey and Stephen T. Ziliak, "The Standard Error of Regressions," *Journal of Economic Literature*, March 1996, <http://www.jstor.org/stable/2729411>.
- Stephen T. Ziliak (2019) How Large Are Your G-Values? Try Gosset's Guinnessometrics When a Little "*p*" Is Not Enough, *The American Statistician*, 73:sup1, 281-290, DOI: 10.1080/00031305.2018.1514325
- Joshua D. Angrist and Jörn-Steffen Pischke, The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con out of Econometrics. *Journal Of Economic Perspectives*. Vol. 24, No. 2, Spring 2010 (pp. 3-30).

Descriptive Statistics and Visualization

- Emmanuel Saez, "Striking it Richer: The Evolution of Top Incomes in the U.S. (Updated estimates)," UC Berkeley, <https://eml.berkeley.edu/~saez/>.
- Robert Pollin, James Heintz, Peter Arno, Jeannette Wicks-Lim and Michael Ash, 2018, Economic Analysis of Medicare for All, Political Economy Research Institute, <https://www.peri.umass.edu/publication/item/1127-economic-analysis-of-medicare-for-all>, Chapter 2 and Appendix 7.
- Robert Pollin, Jeanette Wicks-Lim and Heidi Garrett-Peltier, *Green Prosperity: How Clean Energy Policies Fight Poverty and Raise Living Standards in the United States*, http://www.peri.umass.edu/fileadmin/pdf/other_publication_types/green_economics/green_prosperity/Green_Pro Prosperity.pdf

- Robert Pollin and Shouvik Chakraborty (2015) "An Egalitarian Green Growth Program for India," <https://www.peri.umass.edu/economists/shouvik-chakraborty/item/688-an-egalitarian-green-growth-program-for-india>.
- Ying Chen, 2019. "Renewable energy investment and employment in China," *International Review of Applied Economics*, vol. 33(3), pages 314-334, May.
- Michael Ash and James K. Boyce. "The Toxic 100: Top Corporate Air Polluters in the United States," <http://toxic100.org>.
- Edward R. Tufte. *The Visual Display of Quantitative Information*, Chapter 1: "Graphical Excellence." Cheshire, Connecticut: Graphics Press, 1983.
- Schwabish, Jonathan A. 2014. "An Economist's Guide to Visualizing Data." *Journal of Economic Perspectives*, 28 (1): 209-34. DOI: 10.1257/jep.28.1.209
- Anscombe, F. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17-21. doi:10.2307/2682899
- Fortin, Nicole M., and Thomas Lemieux. "Institutional changes and rising wage inequality: is there a linkage?" *Journal of Economic Perspectives* 11, no. 2 (1997): 75-96.

II. Critical Replication

- King, Gary. "Publication, Publication," PS: Political Science and Politics, Vol. XXXIX, No. 1 (January, 2006), 119-125, <http://gking.harvard.edu/files/abs/paperspub-abs.shtml>.
- Carmen Reinhart and Kenneth Rogoff "Growth in a Time of Debt," *American Economic Review* 100(2) May 2010, pp. 573 – 78. <http://www.aeaweb.org/articles.php?doi=10.1257/aer.100.2.573>
- Thomas Herndon, Michael Ash, and Robert Pollin, "Does High Public Debt Consistently Stifle Economic Growth? A Critique of Reinhart and Rogoff, PERI Working Paper, April 2013, (and Cambridge Journal of Economics, 2014) <http://www.peri.umass.edu/236/hash/31e2ff374b6377b2ddec04deaa6388b1/publication/566/>
- Mark Zandi, 2019, "The Macroeconomic Impact of Senator Elizabeth Warren's Clean Energy Plan," Moody's Analytics, June, <https://www.economy.com/mark-zandi/documents/2019-06-02-Clean-Energy-Economic-Impact.pdf>
- Francisco Rodriguez and Dani Rodrik. Trade Policy and Economic Growth: A Skeptic's Guide to the Cross-National Evidence, in Ben S. Bernanke and Kenneth Rogoff, editors, *NBER Macroeconomics Annual 2000*, Volume 15. MIT Press. January 2001. <http://www.nber.org/chapters/c11058>
- Michael Wachter, Barry Hirsch and James Gillula, 2016 "Wage Comparability of U.S. Postal Service Clerks with the Private Sector, manuscript.
- Robert Pollin, Jeannette Wicks-Lim and Michael Ash, 2019, "An Analysis of Wage-Setting Patterns for U.S. Postal Workers," manuscript, September.

III. Multiple Regression in Price Index Construction

Berndt, Ch. 4.

Dean Baker, *Getting Prices Right*, passim, <http://books.google.com/books?id=p11JdYWSHh0C>.

Charles Schultz and Christopher Mackie, eds., *At What Price? Conceptualizing and Measuring Cost of Living and Price Indexes*, passim, <http://stills.nap.edu/books/0309074428/html/>

David S. Johnson, Stephen B. Reed, and Kenneth J. Stewart, "Price Measurement in the United States: A Decade after the Boskin Commission," *Monthly Labor Review*, May 2006, <http://www.bls.gov/opub/mlr/2006/05/art2full.pdf>

Robert Pollin and James Heintz, "Expanding Decent Employment in Kenya: The Role of Monetary Policy, Inflation Control and the Exchange Rate," UNDP International Poverty Centre, Country Study #6, 2007, pp. 11-15 and 34-37, <http://www.undp-povertycentre.org/pub/IPCCountryStudy6.pdf>.

Marshall Reinsdorf and Jack Triplett, "A Review of Reviews: Ninety Years of Professional Thinking about the Consumer Price Index, 2008, <http://www.nber.org/chapters/c5068.pdf>

See also *Journal of Economic Perspectives*, Symposium, Winter 1998, <http://www.jstor.org/stable/i345523>.

IV. Economic Effects: Comparisons, Controls, and Causality

Disparities and Discrimination, Premiums and Penalties

Berndt, Chapter 5, "Analyzing Determinants of Wages and Measuring Wage Discrimination: Dummy Variables in Regression Models."

Berndt, Chapter 11, "Whether and How Much Women Work for Pay: Applications of Limited Dependent Variable Procedures."

Francine D. Blau and Lawrence M. Kahn. The Gender Wage Gap: Extent, Trends, and Explanations. *Journal of Economic Literature* 2017, 55(3), 789–865. <https://doi.org/10.1257/jel.20160995>

England, Paula, Jonathan Bearak, Michelle J. Budig, and Melissa J. Hodges. "Do highly paid, highly skilled women experience the largest motherhood penalty?" *American Sociological Review* 81, no. 6 (2016): 1161-1189.

Munnell, Alicia H., Geoffrey MB Tootell, Lynn E. Browne, and James McEneaney. "Mortgage lending in Boston: Interpreting HMDA data." *American Economic Review* (1996): 25-53.

Stock and Watson, Chapter 11, Regression with a Binary Dependent Variable.

Health

Rice, Thomas, and Kathleen R. Morrison. "Patient cost sharing for medical services: a review of the literature and implications for health care reform." *Medical care review* 51, no. 3 (1994): 235-287.

Brot-Goldberg, Z. C., Chandra, A., Handel, B. R., & Kolstad, J. T. (2017). What does a deductible do? The impact of cost-sharing on health care prices, quantities, and spending dynamics. *Quarterly Journal of Economics*, 132(3), 1261-1318.

Aron-Dine, Aviva, Liran Einav, and Amy Finkelstein. "The RAND health insurance experiment, three decades later." *Journal of Economic Perspectives* 27, no. 1 (2013): 197-222.

Immigration

David Card, 2009. "Immigration and Inequality." *American Economic Review*, 99 (2): 1-21. DOI: 10.1257/aer.99.2.1 <https://pubs.aeaweb.org/doi/pdfplus/10.1257/aer.99.2.1>

Pollin, Robert, and Jeannette Wicks-Lim. "Did immigrants in the US labor market make conditions worse for native workers during the Great Recession?" University of Massachusetts Amherst, Political Economy Research Institute (2011).

Borjas, George J., Jeffrey Grogger, and Gordon H. Hanson. "Immigration and the economic status of African-American men." *Economica* 77.306 (2010): 255-282.
<https://sites.hks.harvard.edu/fs/gborjas/publications/journal/Economica2010.pdf>

Minimum Wage

David Card and Alan B. Krueger. Chapter 6, "Evaluation of Time-Series Evidence" and Chapter 2, "Employer Responses to the Minimum Wage: Evidence from the Fast-Food Industry." *Myth and Measurement: The New Economics of the Minimum Wage*. Princeton University Press, 1995, <https://www.jstor.org/stable/j.ctv7h0s52> <http://www.jstor.org/stable/2118030>.

Arindrajit Dube, T. William Lester and Michael Reich, "Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties," *Review of Economics and Statistics* 92(4), pages 945-64, 2010. <http://www.irl.berkeley.edu/workingpapers/157-07.pdf>

More technical

David S. Lee and Thomas Lemieux, "Regression Discontinuity Designs in Economics," NBER Working Paper 14723, February 2009, <http://www.nber.org/papers/w14723>

Angrist and Pischke, Chapter 6, Chapter 5: pp.112-138 2SLS and the Wald Estimator; pp.150-158 IV as "Local Average Treatment Effect" (LATE); pp.188-192 Forbidden regressions and other common IV Mistakes; pp.205-216 Weak instruments and the bias in IV, and Ch. 7, pp. 269-283, Ch 8.

Brent R. Moulton. "An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units." *Review of Economics and Statistics*, 72(2), pages 334-338, May 1990, <http://www.jstor.org/stable/2109724>.

Bertrand, M., E. Duflo, and S. Mullainathan, "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics*, 2004, pp. 249-275. <http://www.jstor.org/stable/25098683>

Symposium on new econometric methods, *Journal of Economic Perspectives*, Fall 2001, <http://www.jstor.org/stable/i346045>, including John DiNardo and Justin Tobias, "Nonparametric Density and Regression Estimation"; David Brownstone and Robert Valletta, "The Bootstrap and Multiple Imputations: Harnessing Increasing Computer Power for Improved Statistical Tests"; and Roger Koenker and Kevin Hallock, "Quantile Regression."

V. Macro: Forecasting, Granger Causality, Cointegration, VAR, and the Credibility Revolution

Berndt, Chapters 6 and 8.

Stock and Watson, Ch. 14 as needed and Chs. 12-13 passim.

R. Pollin, "Considerations on Interest Rate Exogeneity," PERI Working Paper #177, http://www.peri.umass.edu/fileadmin/pdf/working_papers/working_papers_151-200/WP177.pdf.

David Gordon, "Must We Save Our Way Out of Stagnation?: The Investment/Savings Relation Revisited" in R. Pollin ed., *The Macroeconomics of Saving, Finance and Investment*, pp. 126-36. <http://books.google.com/books?id=NZUYF-AofyMC>.

Léonce Ndikumana, "Macroeconomic Policy Factors and Private Investment in South Africa," Appendix 2 of Pollin, Epstein, Heintz and Ndikumana, *An Employment-Targeted Economic Program for South Africa*, <http://www.peri.umass.edu/Publication.236+M5adee5a4c07.0.html>

Basu, Deepankar, and Debarshi Das. "Profitability and Investment: Evidence from India's Organized Manufacturing Sector." *Metroeconomica* 68, no. 1 (2017): 47-90.

- Martin Feldstein and Charles Horioka, "Domestic Saving and International Capital Flows," *Economic Journal*, 1980, 314-29, <http://www.jstor.org/stable/2231790>.
- Nakamura, Emi, and Jón Steinsson. "Identification in macroeconomics." *Journal of Economic Perspectives* 32, no. 3 (2018): 59-86.
- Nakamura, Emi, and Jon Steinsson. "Fiscal stimulus in a monetary union: Evidence from US regions." *American Economic Review* 104, no. 3 (2014): 753-92.
- Olivier Blanchard and Daniel Leigh, "Growth Forecast Errors and Fiscal Multipliers," IMF Working Paper No. 13/1. <http://www.imf.org/external/pubs/ft/wp/2013/wp1301.pdf>

VI. Inflation, Unemployment, and Aggregate Macroeconometric Models (4 Classes)

- Berndt, Ch. 10, "Parameter Estimation in Structural and Reduced Form Equations of Small Macroeconometric Models: The Phillips Curve and NAIRU"
- Robert Pollin, "Wage Bargaining and the U.S. Phillips Curve: Was Greenspan Right about 'Traumatized Workers' in the 1990s?" unpublished manuscript.
- Durevall, D., Loening, J. L., & Birru, Y. A. (2013). Inflation dynamics and food prices in Ethiopia. *Journal of Development Economics*, 104, 89-106.
- Robert J. Gordon, "The Time-Varying NAIRU and its Implications for Economic Policy," *J. of Economic Perspectives*, Winter 1997, 11-32, <http://www.jstor.org/pss/2138249>.
- Douglas Staiger, James H. Stock, and Mark W. Watson, "The NAIRU, Unemployment and Monetary Policy," *Journal of Economic Perspectives*, Winter 1997, 33-50, <http://www.jstor.org/pss/2138250>.
- Robert J. Gordon, "The Phillips Curve Is Alive and Well: Inflation and the NAIRU During the Slow Recovery," NBER Working Paper No. 19390, August 2013, <http://www.nber.org/papers/w19390>
- Olivier Blanchard, "Should We Reject the Natural Rate Hypothesis?" *Journal of Economic Perspectives*, 2018, <https://pubs.aeaweb.org/doi/pdfplus/10.1257/jep.32.1.97>
- Michael Bruno "Does Inflation Really Lower Growth?" *Finance and Development*, September 1995, pp. 35-38.
- Michael Bruno and William Easterly, "Inflation Crises and Long-Run Growth," *Journal of Monetary Economics*, 1998, 41, pp. 3 - 26, <http://www.sciencedirect.com/science/article/pii/S0304393297000639>.
- Robert Pollin and Andong Zhu, "Inflation and Economic Growth: A Cross-Country Non-Linear Analysis," *Journal of Post Keynesian Economics*, Summer 2006, pp. 593-614. Also at: http://www.peri.umass.edu/fileadmin/pdf/working_papers/working_papers_101-150/WP109.pdf

VII. Student Oral Presentations of Course Term Papers (3 classes plus extra sessions as needed)

Start thinking about it right away.

Economics 753: Applied Econometrics – Fall 2019

Methodological Issues, Descriptive Statistics, and Visualization

Critical Replication

Multiple Regression in Price Index Construction

Identifying Economic Effects: Comparisons, Controls, and Causality

Discrimination in labor markets and home-mortgage lending

Elasticity of Demand for Health Care

Immigration

Minimum Wage

Macro: Forecasting, Granger Causality, Cointegration, VAR, and the Credibility Revolution

Explaining and forecasting investment

Fiscal stimulus and economic growth

Interest rate cointegration

Triangular systems and VAR

Inflation, Unemployment, and Aggregate Macroeconometric Models

Klein Model I

The Time-Varying NAIRU

Inflation and growth

Student Oral Presentations of Course Term Papers (+ additional sessions)

1	09/04/19	Wednesday	
2	09/09/19	Monday	
3	09/11/19	Wednesday	
4	09/16/19	Monday	
5	09/18/19	Wednesday	
6	09/23/19	Monday	Problem Set 1 distributed, due Monday, 10/7
7	09/25/19	Wednesday	
8	09/30/19	Monday	Reschedule (Jewish holiday)
9	10/02/19	Wednesday	
10	10/07/19	Monday	Problem Set 2 distributed, due Monday, 10/21
11	10/09/19	Wednesday	Reschedule (Jewish holiday)
12	10/15/19	Tuesday	
13	10/16/19	Wednesday	
14	10/21/19	Monday	Problem Set 3 distributed, due Monday, 11/4
15	10/23/19	Wednesday	
16	10/28/19	Monday	
17	10/30/19	Wednesday	
18	11/04/19	Monday	Problem Set 4 distributed, due Monday, 11/18
19	11/06/19	Wednesday	
20	11/13/19	Wednesday	
21	11/18/19	Monday	Problem Set 5 distributed, due Monday, 12/2
22	11/20/19	Wednesday	
23	12/02/19	Monday	Problem Set 6 distributed (see below)
24	12/04/19	Wednesday	
25	12/09/19	Monday	
26	12/11/19	Wednesday	
	12/29/19	Sunday	Term Paper and Problem Set 6 Due